

ActInf Textbook Group ~ Cohort 1 ~ Meeting 10 (Chapter 5 pt. 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_1 | Cohort 1 ~ Meeting 10 (Chapter 5 pt. 1) | 57:06

hello

it is july 7th 2022

it is cohort one textbook group week 10

chapter five

so we're having the first discussion of

chapter five we'll have two weeks of

discussing chapter five

then

in our last two weeks of july

talk about

more general thoughts that we have on

chapters one through five and

look forward to the

second half of the book

um

[Music]

here there's two new

columns

this first one is if you'd like to

join for chapters one through five

cohort two beginning in

september 22

and that will be in parallel with also

this first cohort continuing for

chapters six through ten so just put a

check mark

if you want to stay in otherwise we

won't like include you in that cohort

okay

so

um

[Music]

there's only a few questions prepared

for chapter five

so let's start with any general

thoughts

that people had about chapter five like

how is this chapter similar or different

from the previous ones

and what is the function of chapter five

here

or what what did they think was an

important overall aspect of this chapter

and then of course anyone who can

capture questions that people are asking

here just add them

but other than that it would be awesome

to hear like

what did people think chapter five was

doing or what did they wonder about

about the material or related to it

ali and then anyone else

uh i think chapter five uh

needs a lot of neurobiological

background and without

uh necessary requisites uh it

can be probably daunting chapter

and unlike other chapters which are at

least the previous chapters

uh which concern themselves with

mathematical formalisms uh here in

chapter 5 for the very first time

we encounter some empirical evidence for

uh

or

the empirical uh

reasons behind

uh modeling active inference

as such so

it's a very important chapter in my

opinion but at the same time without any

prior neurobiological background uh it

could be a really daunting chapter as

well

nice

thanks for the summary uh mike

yeah it was an interesting choice to go

all the way to the human brain to

illustrate how these concepts would map

to biological systems

as opposed to

starting with more simplistic organisms

and kind of working up from there

yeah very nice point

any other overview thoughts and then we

can just like scan through it and look

at the order of a few things and then

continue to write questions

again like as you're hearing them or

like as you're just wondering

but let's um

yeah hear any other general thoughts if

anyone wants to raise their hand and

then look at the order

of the chapter

okay

well the chapter is titled message

passing in neurobiology

message passing was brought up

in

a previous

chapter

in this box 4.1 in a

formal

yet slightly non-standard

notation without examples just generally

the quote

anyone have any thoughts on

jeff

yes i kind of disagree with the quote

i i think uh recently we've we've

learned that the plants actually uh
sense and act on their their their uh on
what they they detect in the environment
for example
uh certain trees when they're being
munched on by um
say like giraffes
will emit uh
molecules that will disperse and other
trees that would that
sense these uh this essentially an alert
signal will start producing uh compounds
in the leaves that make the leaves taste
very bitter and thus discouraging uh
foraging

[Music]

and so forth and so on and also we you
know
i think it's clear that plants sense and
act
and respond to their environment and act
in order to maintain their integrity or
as individuals or as uh as groups
in the example i just
mentioned
nice agreed
oh also the quote might have like a
little bit of hyperbole slash
comedy like two types of animals
plants
so
it's about
nervous systems ali
uh yeah and uh one other thing about
rodolfo linus is um in one of his most
famous books eye of the cortex from
neurons to self
um actually he describes thinking

uh as an internalized movement uh and
it's a very interesting way to describe
uh thinking in my opinion and uh it
directly relates to this active
inference uh framework so

uh

yeah putting uh his quotation here might
be quite relevant

cool and then i added a 2017 paper with
fristen

and calvo about plant and predictive
processing framework

and then here's a like a non-active
paper but about more on the behavioral
side

which is kind of the view from the
outside

without denying the view from the inside
as well

but

it's just related to behavioral modeling

okay in the

uh introduction

they write let us take a step back from
the technical material of chapter four

which as um

ali and others have reminded us was like

even at the beginning was like you can

bypass this chapter if you don't want

technical detail

so

maybe somebody had read it even one two
three

five

and maybe they didn't look at the

appendix

turn to the process theory's

accompanying active inference

drawing a distinction between a
principle
and a process theory
does anyone have any thoughts on that
like how as they're reading the book are
they thinking about
where free energy principle is related
to active inference
how does introducing that
kind of broad
map territory slash
principle process
distinction here help transition the
book towards focusing on these empirical
cases
ali with a hand raise or anyone else
okay just one
yeah yeah go for it
yeah
sorry just one thing i forgot to mention
about that quotation
is uh
as an interesting research has been
emerging uh that's
uh kind of
um
so consciousness to plants as well uh
one of
the
articles uh
one of the early articles for that is
insights into plant consciousness uh so
uh in fact
we don't necessarily uh
limit it to
just human or even animal consciousness
and we're beginning to relate
consciousness or

any other aspects of nervous system to
plants as well so just one additional
remarks about that previous code sorry
cool
yeah and the mathematical like framework
for behavior and consciousness that
are integrative
will allow comparisons of different
kinds of behavioral systems and that
like leads to people wondering about
consciousness and all those other areas
um
so
mike and then anyone else
yeah so i'm tying that back to
the part of the chapter that you're on
previously in terms of moving from
principle to process
and and you had highlighted that
the latter allows us to develop
hypotheses that are answerable to
empirical data
so i think this is um
the the first part of
the first instance in the book where
they are
um
making
a serious effort to tie out to real
world observations as opposed to saying
here's a a framework for how things
might work and how the pieces might be
connected now we're going to take that
framework and think about uh
can we
identify
things that we see in real systems in
this case in the human brain

and tie those back into the ideas that
we put forth previously
thanks totally agree it's like
partitioning off
the principles from the earlier chapters
like in principle those two sides of the
equation are equal because of how
the axioms of math are or like in
principle one could do this with a
bayesian graph
and then that is like it's like in
principle you could do a linear
regression with a least squares error
minimization nobody can make any data
set that says that you can't do that
and then there's how that least squares
regression works
on the empirical data
so they're saying for chapter five we're
going to
look at
with all the architecture developed
a few specific neural systems
and talk about how they're
the best
current
uh understanding that they have of how
these neural systems can be
modeled with the models of the uh
structures like we've seen before
and they're
saying we're not trying to write a neuro
textbook here
but to touch
kind of like with both feet the
empirical modeling
and then the systems that they are going
to

model

are like

effector systems so motor control

subcortical structures like the thalamus

and basal ganglia

the which are so these are more like um

functional outcomes of the nervous

system

specific anatomical structures of

interest

modulation of synaptic efficiency or

efficacy

which is like a micro anatomical or like

a neurophysiological

mechanism of interest

if and something that is an important

like leverage point and something that

is being modified and found to be

changed in different situations

and then

um

the relationship between decision making

and movement generation

which is also something that has come up

in those um 46 live streams

on like motor active inference decision

making active inference

so what did

uh well let's check if there's any

questions written

okay

um what could anyone say about

the cortical layers

or what was in figure 5.1

mike so

i was just observing that

um

in

in the start of chapter four they said
well you can skip this chapter if you
don't want to get deep into the math and
then here we are at section 5.2 in
chapter 5 i think and they refer back to
the message passing
structure that was set up in chapter
four

yes

good point

and it's like well if you can skip the
uh details of the formalism to look at
the examples or you could skip the
examples

any linear

text 140 characters or textbook or you
know live stream or anything it's always
going to be some sort of linear
presentation

question

and with a densely
linked

and interdisciplinary area then like the
linearity becomes challenging
so maybe with more

you know with some notebooks or some
other rendering or
educational way to

have the ordering or have the of course
the connections between sections be a
little bit different

um the first

system that they're focusing on is the
cortical column

does anyone have like any
thing to add on

cortical layers or the role of the
cortex

even if it's just something that they've
heard about
well i think that
this um
numenta line of ai
and um many of the other oh i'll lea go
for it yes
i just i want to say that
uh well
for the past
and couple of decades or probably
several decades uh cortex uh has
traditionally been thought of as the
seat of both consciousness intelligence
and also of course cognition but
recent studies has shown that especially
studies such as mark some
research has shown that in fact
consciousness actually stems from uh
literally the brain stem not uh
necessarily the cortex so uh there are
some challenges about the exact function
of cortex
and
the debate goes on
nice okay
here's a little bit on the cortex
so it is the outer frontal part of the
mammalian style brain
insects don't have
cortex
um
though they share a lot of the same
architectures and so on so this is like
neuron tracing
and there's kind of two levels of
organization that people are modeling in
the cortical models the first is like

based upon the
histological observation of the layers
these are six layers of like
cells and there's like a lot of
development and all
different functionality between the
different layers
um one important thing to note is like
it's not that they're like the layers of
a bayesian hierarchical model they're
not like it's not like each one is like
up and down up and down so this isn't
like a predictive processing
architecture happening
simply with these six layers this is not
a six layer model
in fact they're connected to each other
sparsely within a column as it's shown
here in 5.1
so that's like the micro anatomical
structure of the cortical column
and it's called a column because it's
like the layers are also arranged in
this repeating way
and there's just like
a lot of different um
layouts
for different
regions of this
cortical tissue type
and in certain situations it is
organized
like
linear corner structures
which is often seen as like one of the
evolutionary
local maxima
of like

depth of processing
and sparsity of connections laterally so
those are the two levels of analysis
that are being and as ally mentioned
like people yes i've gone to the
consciousness angle or they've just
argued that this is kind of like the
massively paralyzable but also like deep
recurrent
contextual
etc the anagrams like the patterns that
can be activated
the dimensionalities potentially super
high so it could play a role in memory
signal processing
and the two levels that people look at
the connections between
like anatomically by taking these images
or by um doing like stainings and stuff
um anatomically and functionally is
within the column where there's certain
kinds of relationships and then across
columns where there are certain kinds of
relationships
so
um
any thoughts or questions on that but
like that's just like why the cortex is
being studied and that's like a little
bit of how like the neuroscientists
have been approaching this
and why this is one of the important
systems with a lot of theoretical
modeling and empirical evidence and like
relevance as well
um jf
i remember many many years ago
taking a course in

physiology on the visual cortex
and you know the whole notion that the
brain
has some kind of structure
um always i thought was fascinating
because i i feel that
the brain is a huge hack
patches upon patches upon patches
brought by evolution and that
there would be some structure in it is
is
is uh
is interesting but within within those
structures there's as those this picture
shows this it's very messy i would never
implement a system with this kind of of
wild interconnectivity
so
i'm very
curious about
how much we can
infer
from
what is really a
mildly structured but highly messy
architecture in terms of the the process
theory that we're trying to uh to
elucidate
from anatomy and the functionality of
the brain
nice thank you mike
yeah that makes me think that some of
that messiness is
in support of redundancy and so
how do we think about redundancies in
systems in the context of active
inference
yeah and like neural systems of course

have amazing
in some dimensions ability to have
redundancy other times seemingly there's
points of failure jeff
okay okay all right so just
um on the left side i'm sorry oh
yeah sorry
uh on the comment about redundancy i
agree but it's worse than that it's it's
uh every part
often plays
many different roles and and participate
in different functions which makes it
quite a tangle
uh in terms of uh functional analysis of
parts of the brain what they do and what
not
so five one is showing
um a
more like pure
base graph
um
with the annotation of the kinds of
cells
that are in the anatomical layer so this
is like
a visualization of the anatomy based
upon the regions that it connects gets
inputs and outputs from
and then the micro
anatomy labeled according to cell types
then
with a similar but not exactly
concordant there's not a statistical
test proposed for how concordant theos
are it's not a statistical argument it's
not exclusive of other framings of what
this anatomy is doing

here are labeled with different parameters from the base graph architectures that have been explored in the previous chapters compatible with message passing implementations that other papers have addressed that aren't here these are like architectures that recapitulate some of the functional and macro structural and microstructural aspects so this is like computational neuroscience and it's using these models that actually have an underlying structure that is providing utility and interesting reframings of the anatomy and as the model um statistical architecture begins to recapitulate aspects of this of the cyto architecture there's like no specific threshold where it's like oh well now it's like an argument that it's you know doing that function like people get tantalized by it having some oh we made this network structured like a brain or we made this network structured like a mushroom or we did the ant colony optimization so just because like this works for what it does doesn't mean this works

for what anyone wants it to do
it's just in this huge space of
bio-inspired model architectures
in um
five two they move a little bit from the
consideration of the columnar
architecture which are cell populations
to
um looking now like still the
superficial to deep are the six layers
of the cortex
but now a few of the variables are
highlighted and they're relationships
and so
the um
um
ascending and descending prediction
error in a predictive processing
framework which is like often
graphically seen going up and down
bottom up and top-down effects
here
is being shown in the context of the
columnar
cortex
where like errors
from one
like let's just
go to where they
um
the
e's are the errors
and the g's are predictions
and um
i guess we can find what μ is but
basically like this architecture within
and between
columns

could be like this being like
the base variable being estimated and a
hierarchical nested model of that and a
hierarchical model of that
so there's like a generalization
statistical architecture
that has compatibility with the neural
architecture of the brain
and it does like a pure statistical
function
and that pure statistical function has
like resemblance with some things that
the brain region is believed to do
so that
allows this statistical model to be
explaining and pre predicting
and useful
about the brain which is like what
computational neuroscience does
so that was the first
one of the systems that they focus on
then
oh yeah i guess this was
um then the factor systems okay let's
see if we can kind of quickly look at
all the systems
what
did anyone see or what would they like
to say about figure five three
neuroanatomy associated with active
inference in modulating spinal motor
reflexes
um
okay
so five one was the columnar
architecture
and that was described in terms of what
cell types were in it

here they're going to pick up on one of those cell types which is like the computational role the expectations encoded by this cell type that's found in this layer of the motor cortex and so different regions of the cortex have different apparent functions asterisk asterisk but there's like a visual cortex in a motor cortex and so on so this is in a motor region of cortical architecture that plays a certain biological function the expectations are subtracted from the incoming proprioceptive input in the horn of the spinal cord this is like lateral cut through a spinal cord the horns are like the butterfly parts and so there's a differencing between the incoming intensity of proprioception and the brain expected expectations and then the error drives muscle activity in the direction of suppression so the motor activity is like reducing the divergence based upon the sign and the intensity of the compromise between the brain's expectations potentially arising within like layer 5

from these bet cells
and the incoming sensory information
which can be modeled as like why
i think μ is probably the mean of that
expectation
um
i
don't know what part it was mentioned
um and then here's the
expectation the the
difference between these two it's like a
causal system flow but like a little
different
you're getting a 5 out of 10 pressure
you're expecting a 5 out of 10 pressure
the motor reflex is not engaged
you're getting a 7 out of 10 pressure
you're expecting 5 out of 10
there's like a negative 2 difference or
you know whichever way you want to be
subtracting there's like a 2 or a
negative 2 difference
and then that triggers like some motor
neuron units to fire
and then that
proportionally brings
the um
expectation
into alignment with the measurement
um
they're gonna just like that example
hinged from
a cortical layer and cell type
here is going to be hinging on the
cortex and looking to an output region
of the cortex
which is the striatum
and then they're going to do a more

detailed model of a basal ganglia
because it's been modeled a lot more and
then also have like just a textual
treatment of the thalamus
so
there's multiple papers on
dopamine
and active inference
and so um
from
2012 and then
2015
like
this is a really interesting
of research
with dopamine
because
dopamine is like classically
described as the reward molecule
more reward more dopamine lower reward
lower dopamine
um
so
isn't the whole
meme inactive that we replace the
reinforcement reward learning paradigm
with a
different
imperative that has elements that
reflect pragmatic value
but that isn't
the um overarching
uh
imperative for policy selection and so
these papers
show how like
in the context of a generative model
with expectations and preferences and

dope means signaling
something related to the predictive
processing of stimuli
under a mildly optimistic world model
good events and surprising events are
still associated with dopamine release
but
and things that are going worse than
expected are still associated with like
dopamine drops
but um
that actually is like a more unified way
to talk about
dopamine's
empirical outcomes
as opposed to the
reward learning which has to
hypothesize all of these like
modules that translate different kinds
of things into reward jessica
yeah no i think similarly to these i
don't remember where i read this but it
isn't like
the brain like in regards to like the
dopamine comment it's like it's also
like
the anticipation
on
of doing something and like the fact
that you're gonna get that dopamine kick
and
like you get basically you get a release
by anticipating something and that maybe
a little bit connects also with the
active inference the fact that you are
anticipating something
you know it's like you expect that to
happen and so you get that release of

the dopamine in anticipation of the event happening so i don't know much but like i'm thinking it could be connected with having like a predictive unexpected free energy or something nice good insight rohan yeah uh i'm not uh well i'm not a biologist but okay so i just wanna clarify there are other systems in the body apart from just the brain alone right so you have like the autonomous uh or what is it parasympathetic nervous system that controls your heart rate uh your respiration and other things and they and also you have stuff like adrenaline and not adrenaline uh that do produce responses and other hormones also that produce responses like uh okay for hunger or satisfaction and we we basically have figured out how to modulate these things to solve modern issues for things like obesity or something like that and there are treatments available that modify these hormones levels right so is it is this not a very limiting perspective to look at dopamine alone as the only thing and my other question would be how does that convert into say motor commands so i just increase dopamine levels without any

stimulus whatsoever

let's say someone's taking a drug

it would

you say that that does not necessarily

translate to a motor command right so

how does it

uh do this is there's like a specialized

system

that figures out okay so here's uh

here's the previous action here's

increase in dopamine and here's how we

modulate to further increase dopamine

and

where exactly do you set limits on this

uh process

okay the two questions were about

interacting physiological systems and

the second one was about dopamine means role

in motor selection um blue go ahead if

it's related

um or if you want to address anything

yeah no i just wanted to respond like

the dopamine response is like so um

there are a lot of interactions that are

happening in the nervous system dopamine

is one of the best understood i think

it's one of the ones that we've studied

for a long time

um and the if there's like an overload

of dopamine

the action can actually be non-specific

and that's what we see in parkinson's

right so it's like an overload of

dopamine into that

system and and it fries it out and then

you get non-specific

um

action

in the nervous system

motor action the tremors and stuff

um yes

good call and that like question about

um

like um non-purposeful behavior and

dopamine on the positive and on the

negative like failure to engage

initiate action

relating it like catatonic behavior and

then like hyper motor activity

associated with um repetitive motions

of various kinds

like dopey dopamine is known to be

different in people who are experiencing

that differently and modified by drugs

that target that like system so um

but to the first question about like

pluralism with respect to physiological

systems

for sure this model does not exclude

that

in fact its flexibility

is its advantage because the inputs to a

brain region now we could say well at

this time scale it's electrochemical and

at this time scale it's going to be just

um you know noradrenaline and someone

says well

how could you only model noradrenaline

say okay let's look at the model with no

adrenaline and this other one and an

unmodeled component

so

it will allow integration formally

of

physiological mechanisms that genuinely

interact and as to why people write like

specific papers about specific hormone

systems it's just like

what

is chunked off in that fractal and then

also one important note is like um

columbo and wright

have written about how like we don't

need a monolithic theory of any

neurotransmitter or any brain region

there might be like features or

components of dopaminergic systems

that it just explains and predicts that

they do reward

taken lightly that the explanation is

just of course our explanation of a

biological system

um

and also you raise like beyond the brain

there's also glia and other cell types

so yeah

the cool thing would be a framework

where we don't a priori exclude factors

that are important

um but just to kind of go quickly

through

the example and get to that motor

um

action

so this is an output from some putative

processing that happened in the cortex

and now it's projecting into some

dopaminergic regions

in the subcortical area

like the basal ganglia

um

it's going to be a model involving

outcomes o

the difference between the preferred and

the expected outcomes
the sigma squiggle expected free energy
evaluated for a policy and the um
posterior over policy is the bold pie
um the figure is like shown after a lot
of the description
but um
here is where they refer to the high and
low dopamine state so also like the the
movie slash book awakenings
is about um people who have like um
atypical dopamine signaling and then
they're given l-dopa and they like come
and like get re-invived with life and
stuff um rohan
yeah so
well this is
a little more general but okay when so
we have the marco blanket and there are
some internal states and the markov
blanket
basically connects us to the external
states right
uh or
samples from the external states and
then
modulates the internal states
accordingly
uh okay so
my question is i think if the if the
first imperative is survival
and the first imperative is to keep
uh this system in homeostasis which is
composed of multiple
interlocking and interacting complex
components right
how exactly would you translate
like these multiple signaling

modalities with multiple hormones
into maintaining homeostasis with
something like free energy
because some of these things are
compatible entirely that like
you would have to have some sort of
translation mechanism right so
let's say that adrenaline is jumping
really high
and into dangerously high levels but we
only know it's dangerous because
there is some system that is able to
predict that okay it should not go above
this level because it's going to screw
up something else
in the system right
right
multi-scale optimization and not over
optimizing one lower level parameter at
the cost of some higher order parameter
yeah but then you
come to the curse of dimensionality so
how many
uh different things are you going to
take up before it becomes completely
useless
so absolutely good question empirical
question
yeah about what computational hardware
what measurement data sets what sparsity
of system graph
what extent of application of heuristics
how accurate you need that model to be
in the empirical setting um none of it
in principle um
[Music]
addressed by just the equations all of
it related to how they're actually

implemented at what scale
just to quickly complete through here
though
this is about the depletion of dopamine
is being observed
um creating this akinesia failure to
move
and exogenous dopamine promotes
impulsive behaviors of multiple
different kinds
um
this is what
it looks like
dopamine as again stated here is
balancing modulating the balance as a
neuromodulator
between inferring what to do and what
not to do
and so these
two graphs
both of them are getting
outcomes
from the cerebral cortex
of the observations expected under a
policy so those could be considered like
predictions about at future sensory
states
here
um and the gamma is this uncertainty
that's being represented by dopamine in
in a like
in this uncertainty function of dopamine
um
[Music]
in the direct pathway the policies and
the difference between the preferences
and expectations sigma squiggle with a
tilde

are being
evaluated as part of the expected free
energy of future policies
and that is resulting in the selection
of policies
so that could be like thinking through
all the chess moves and then picking the
one with the lowest expected free energy
here
the output in the indirect pathway feeds
into
the prior
vector e
habit
and then that
influences policy
through this
acting anatomically proposed to this
other region not as not defined with a
specific variable so again this is not
like saying like necessarily even just
what it's doing this isn't the full
model the papers read more but that's
like with the two pathways of action
selection
and it's compatible with the empirical
evidence from pathology and from
pharmacology and genetic studies and
animals other than humans
and like
this is um
how it could be modeled with a base
graph that recapitulates some of the
architecture and the function
then
they describe the roles of some
different neurotransmitters that have
been

modeled in

active

um

and like what kinds of phenomena

those measurements have been used

in models

um

and then

a one last important part

it was in the section with the basal

ganglia

that was uh

talking about this so they show the

graph of the basal ganglia and then only

in this last

paragraph did they describe the thalamus

so just say we're just going to do one

paragraph on it

but basically the thalamus has these two

divisions

and then they provide the um

interpretation that these two divisions

of the thalamus

could reflect

first and second order statistics

on other

types of senses and policies

so that's like

um

[Music]

a prediction a unique prediction that

potentially modeling uh observed neural

responses or the bold signal from um

like the blood oxygen use

in that region

you'll make a better model if you fit

into these categories rather than reward

or rather than familiarity or any number

of other criteria so just simply fitting
better is not the whole substance of
validating the formalisms described
earlier
but
it's part of the empirical grounding and
examples of like unique predictions and
explanations that are provided by active
inference
and that but that was the thalamus part
and then um
in this very short section 5 6
they
talk about how
a lot of our interfaces
are continuous
like sensory apparatus and motor
apparatus have continuous aspects
but then
in the cognitive and decision-making
domains
there's often much more discreteness
even if there's multi-scale discreteness
resulting in like a very finely graded
continuum of alternatives
still there is
um discreteness involved
and this was explored
a lot more fully
in the 46 live streams
active inference models do not
contradict folk psychology
where they
really clarify this is motor active
inference in the continuous domain this
is decision making active inference in
the discrete domain
here's what a hybrid model looks like

blue
oh do i still my hand up sorry oh spin
up
yeah so then they just basically say
um some of these models dealt with
continuous phenomena
like specifically the reflex arc example
hashtag active inference in continuous
time figure 4 3 etc
and then other models presented in this
chapter
are compatible with discrete
alternatives like discrete variables for
example policy selections
if there's two model policy outcomes
then this is like a discrete model or
the observations could be happening
through discrete time so it's like
different opportunities for discreteness
and
continuous models and variables
to be integrated
[Music]
in
these architectures
then
in summary
they're
outlining the points of connection
between the message passing
implied by the generative models of
chapter four
which we asked you not to read
gently
and
the neurobiology of inference action and
planning
and then here's going to be like a

summary figure
for the chapter
the top row is showing some
computational motifs
that are plausible or compatible with
different like subsets of cells and
regions in cortex
the bottom region are showing some
extracortical
structures
on the left is the basal ganglia
shown from like one of the previous
figures
so it's a subcortical structure within
the brain
here is a spinal cord
um cross section
and one can imagine that like other
structures within the brain and outside
the brain
would be amenable to also this kind of
graph
and they're just doing like kind of
linking now even the models together
and then
showing the difference in planning like
on the left side the basal ganglia is
getting this preference difference
is connecting to the expected free
energy so
future policies are being evaluated
policies are being evaluated on the
basis of their future ability to align
with preferences
in this pathway
here that's planning
habits
are when the not the preference

versus

expectation um difference

but

the outcomes themselves are going to e

and

habits are being followed which could be

seen as multi-scale habits

and that's influencing policy

and then here's where that kind of

continuous decision and modulated

handoff between more planning like and

more habitual

is being related

to the motor implementation

of the expectations

of the proprioceptive loops

so like i want to be sitting still

and then all of your muscles in a

healthy situation are like not moving

versus like i want my leg again speaking

like loosely slash

you know with it i want type word but

just like i want to lift my leg

that model

the expectations have to be updated

to allow the leg to realize being up

using a non we don't say well it's more

rewarding for the leg to be up

we can just say this is

a computational architecture that

facilitated that

and inc and generalizes to like

multi-step planning

and then there's been multiple papers

2017 and beyond

that like summarized and elaborated on

different aspects of this

more complex continuous architectures

more complex motor
discrete interfacing more complex
decision making active inference of
course rohan
uh yeah i'm just wondering because a lot
of this has to do with high level uh
more abstract behaviors or kind of
abstract behaviors right
so is there any attempt made
to apply this to something simpler like
the autonomous uh
nervous system because not every signal
goes through to the brain right so that
we just go to the spinal cord and we
know this because if the spinal damage
you can't
sometimes people can't walk or they have
trouble rebreathing and so on because
not
so or in other animals like the
cuttlefish for example that
changes in the pigmentation of the
skin
is not dependent on
brain movement uh on brain activity it's
just
autonomous
right so it does this war has there been
any attempt made to apply
this to much more
uh
you know those kind of systems
um well
uh how simple or complex the system is
is
kind of a it's kind of like how
complexly we approach it because people
have been modeling single cell

um graphs

if that could be seen as the simplest
electrochemical decision maker as well
as um morphological computing
and yeah there's probably a paucity of
published models

on

autonomic nervous system active
inference models

so there are many opportunities because
they cited a huge amount of the neural
empirical work here
with the goal being to summarize sort of
the tip of the iceberg

but

like there aren't

um

models

like this for every single function or
region or or anything

nor even if there were would it be the
end of research in that area

so yeah there's a lot of open space to
build specific models which is exactly
what section six and beyond and part two
will take us to which is to go from like
identifying it we can quickly find out
whether it's been published or not
um and then build that model with the
inputs and outputs that people are
seeing as relevance

rowan

uh yeah i'm sorry for

monopolizing this

question time i'm sorry so i'm just
curious if this has been applied to
non-biological systems in any way
because that seems to be something that

we could easily do right
because we have plenty of data
right now okay so can we is there
anything like a better
vision model than a convolutional neural
net for example or
some sort of swarm robotics
uh
attempt using something like this
or even like uh
game playing or something because i
would be interested to
read
many many of these things have been um
sketched or are in progress we have our
project ideas
in these like last minutes
there's
a lot of
room for
building projects and there's a huge
value to creative
ideation of like areas where it could be
applied
and also value for checking the
literature for
where it might have been like a a model
could be reused like a visual foraging
model
that was from about cultural acquisition
of pattern preference
was repurposed for our aunt stigma g
model
just by adding a pheromone trace
so there isn't like a cookbook on what
is out there or how models could be
modified
we hope that with active block for ins

and
documentation it'll be clearer and
clearer for people
to build generative models and compose
them and explore different modules
so it should just be done
yeah okay thank you
any last thoughts or questions on
chapter 5 and then next week we'll
return to the more specific
questions
ali and then anyone else would like
a closing thought on chapter five
yeah i just want to mention um thomas
parr the
speech uh in the i think it was a
few weeks ago in the center for
cognitive neuroscience berlin
which i'm putting the link here in which
he explained most of these topics and
some more
uh in perhaps a more accessible way so
if anyone wants to
yes that's it if anyone wants to watch
it i think
uh that can help in understanding the
content of this chapter
nice thank you
yep
looks like a good one to watch and using
like the same figures
so that's pretty maybe we could you know
annotate like the figure to say at this
time step here's where it's described
um
okay
so those who want to um feel free to
stay in this room for dot tools

otherwise thanks for joining and see you
next week